New York State Crime Profiling

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***Abstract*—Understanding county-level crime patterns is important for identifying regional differences in crime burden and supporting data-driven public safety analysis. This project examines crime behavior across New York State counties using crime and population data spanning from 1990 to 2024. Since the original crime records were available at the agency level, the data was aggregated to county-year totals and standardized using population-based crime rates. The study investigates both violent and property crime structures. Correlation analysis and variance inflation factor (VIF) were used to examine relationships among the crime-rate variables and assess multicollinearity before model development. K-Means clustering was then applied separately to violent and property crime variables to identify counties with similar crime characteristics, while Principal Component Analysis (PCA) was used to visualize and interpret the cluster structures. In addition, Logistic Regression, Decision Tree, and Random Forest models were developed to classify observations into High, Medium and Low crime categories. Model performance was evaluated using accuracy, precision, recall, F1-score, confusion matrices, and feature importance. The results show that robbery-related variables play an important role in distinguishing crime categories and that Logistic Regression achieved the best overall classification performance, followed by Random Forest, while Decision Tree provided a more interpretable but less accurate structure.**

***Index Terms*—Crime Analysis, Correlation Analysis, Variance Inflation Factor, K-Means Clustering, Principal Component Analysis, Logistic Regression, Decision Tree, Random Forest, New York State, Data Mining**

1 Introduction

Crime is not distributed uniformly across regions, and its intensity often varies according to population size, location, and offense type. Studying crime at the county level provides a clearer understanding of how different areas experience violent and property crime in different ways. In New York State, counties differ widely in terms of urban concentration, population size, and reported crime burden, making county-level analysis useful for understanding regional variation in crime.

County-level crime analysis is important because raw crime counts alone do not always provide a fair comparison across areas. A county with a larger population may naturally report more incidents than a smaller county, even if its overall crime burden is not proportionally higher. For this reason, population-adjusted crime rates provide a more meaningful basis for comparing counties. Such an approach helps identify whether a county has relatively high, medium or low levels of crime after accounting for differences in population size.

Understanding county-level crime patterns is also useful from both an analytical and policy perspective. Identifying areas with similar crime structures can support more targeted interpretation of regional crime behavior and help reveal whether certain counties share common violent or property crime characteristics. Instead of treating crime as a single overall phenomenon, separating it into violent and property crime categories provides a more detailed picture of how crime varies across locations and over time.

Data mining methods provide useful tools for uncovering these patterns. Clustering methods can group counties with similar crime characteristics, while classification models can help determine how well crime categories can be separated using selected variables. These techniques allow the analysis to move beyond descriptive summaries and uncover deeper structure in the dataset. In addition, feature significance analysis helps identify which variables are most strongly related to crime variation and whether any predictors contain overlapping information that could affect the stability of the models.

This project focuses on county-level crime pattern analysis in New York State using crime and population datasets from 1990 to 2024. Since the original crime data was reported at the agency level, the records were first aggregated to obtain county-level annual totals. The crime data was then merged with county population data so that crime rates per 100,000 population could be calculated. This preprocessing step helped create a consistent and comparable dataset for further analysis.

After preprocessing the data, the study applies feature significance analysis, clustering, and classification models in order to understand county-level crime structure and compare the performance of different machine learning approaches. Correlation analysis and Variance Inflation Factor (VIF) were used to examine relationships among the predictor variables and address multicollinearity. K-Means clustering and Principal Component Analysis (PCA) were then applied to identify and visualize county groupings based on violent and property crime patterns. Finally, Logistic Regression, Decision Tree, and Random Forest models were used to classify observations into **High, Low,** and **Medium** crime categories.

Overall, the purpose of this study is not only to compare model performance, but also to gain a deeper understanding of how crime differs across New York State counties. By combining statistical analysis, clustering, dimensionality reduction, and classification techniques, the project aims to provide a more structured view of county-level crime behavior and highlight the variables that play an important role in distinguishing crime patterns across the state.

2 Objective

The main objective of this project is to analyze county-level crime behavior in New York State and identify meaningful patterns using data mining techniques. The project specifically aims to cluster New York State counties based on crime patterns and examine similarities and differences in violent and property crime across counties.

The second objective is to build classification models that can categorize observations into **High, Medium,** and **Low** crime classes using county-related and crime-related variables. This makes it possible to compare different supervised learning methods and determine which model performs best for this classification task.

3 Overall Crime Comparison

Before model building, an overall comparison of county-level crime rates was examined. The descriptive analysis showed that violent crime was more concentrated in a smaller number of counties, especially major urban counties. In contrast, property crime was more widely distributed across multiple counties. The year-wise comparison of maximum index crime rate also showed that the county carrying the highest crime burden changed over time while the overall maximum rate declined.

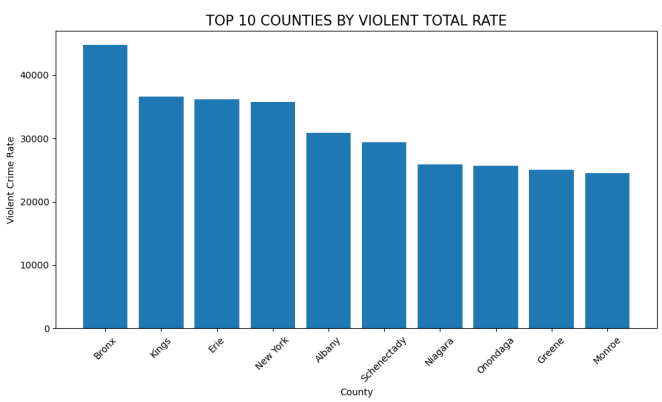

Fig. 1: Top 10 counties with the highest violent total crime rate.

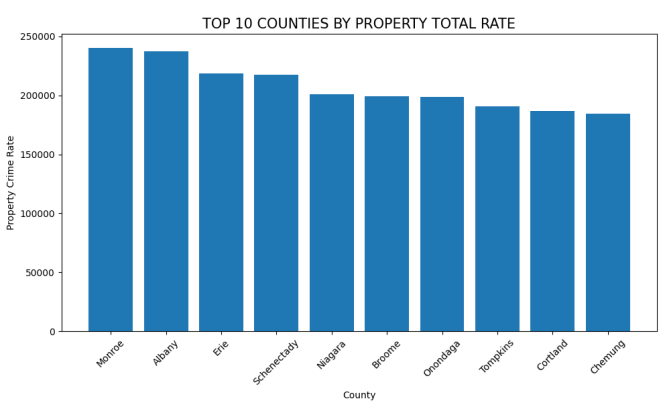

Fig. 2: Top 10 counties with the highest property total crime rate.

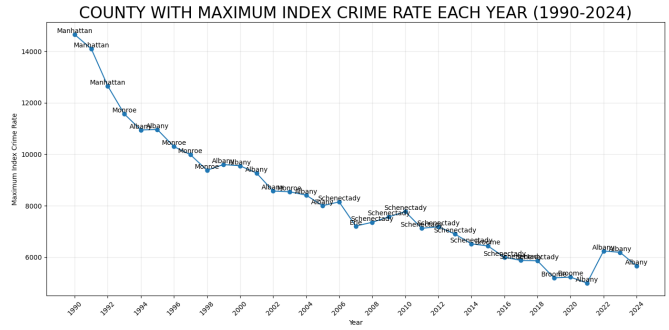

Fig. 3: County with the maximum index crime rate each year from 1990 to 2024.

These observations provided useful background for the later clustering and classification stages, since they showed that counties do not follow one uniform crime pattern.

4 Dataset Description

This project uses two main datasets: a crime dataset and a population dataset. These datasets were combined to study crime patterns across New York State counties and to support clustering and classification analysis.

4.1 Crime Dataset

The crime dataset contains annual crime counts for counties in New York State. It includes major crime categories such as Index Total, Violent Total, Murder, Rape, Robbery, Aggravated Assault, Property Total, Burglary, Larceny, and Motor Vehicle Theft. Since the original crime data was available at the agency level, it was later aggregated to obtain county-level yearly totals. The analysis covers historical data across multiple years, from 1990 to 2024.

4.2 Population Dataset

The population dataset contains yearly county population estimates for New York State. This dataset was used to standardize the crime counts and was merged with the county-level crime dataset using County and Year as the common keys.

5 Data Preprocessing

The preprocessing stage transformed the raw crime and population datasets into a format suitable for county-level analysis. Because the original crime dataset contained multiple agency-level records for the same county and year, the data first had to be aggregated at the county level. Missing values and duplicate records were then handled, the datasets were merged, and crime rates per 100,000 population were calculated.

5.1 County-Level Aggregation

The original crime dataset contained multiple records for different agencies within the same county in a given year. To obtain one record per county per year, the data was grouped

by County and Year, and the crime counts were summed. This produced annual county-level crime totals for each crime category.

5.2 Handling Missing and Duplicate Values

After aggregation, the datasets were checked for missing values and duplicate records. Null values were removed where necessary, and duplicate entries were eliminated to improve consistency and avoid errors in the later stages of analysis.

5.3 Merging Crime and Population Data

Once the county-level crime data was cleaned, it was merged with the population dataset using County and Year as the common keys. This step combined annual crime totals with the corresponding county population values.

5.4 Crime Rate Calculation

After merging the crime and population data, crime rates per 100,000 population were calculated to make county comparisons more meaningful. The rate was computed using:

$$\text{Crime Rate} = \left(\frac{\text{Crime Count}}{\text{Population}}\right) \times 100000 \qquad (1)$$

Using this formula, the following rate variables were created: *Index_Total_Rate, Violent_Total_Rate, Murder_Rate, Rape_Rate, Robbery_Rate, Aggravated_Assault_Rate, Property_Total_Rate, Burglary_Rate, Larceny_Rate,* and *Motor_Vehicle_Theft_Rate*.

6 Feature Significance Analysis

6.1 Correlation Analysis

To better understand the relationships among the selected crime-rate variables, a correlation heatmap was created. The heatmap showed several moderate to strong positive relationships among the predictors. In particular, *Larceny_Rate* and *Burglary_Rate* showed a strong positive association, while *Robbery_Rate* and *Motor_Vehicle_Theft_Rate* also showed noticeable positive relationship. The analysis suggested that some predictors carried overlapping information and therefore required further multicollinearity testing.

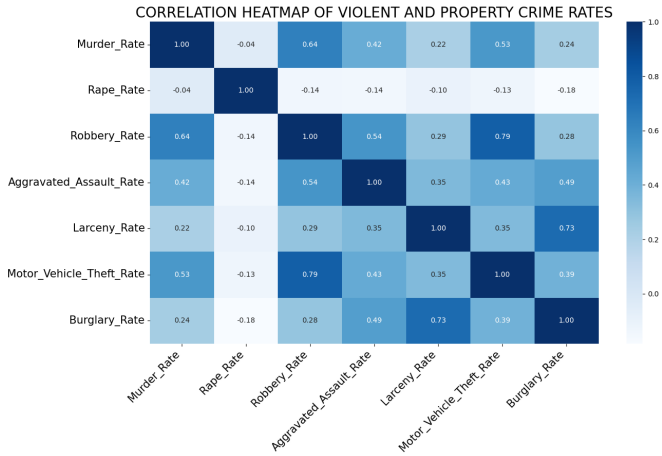

Fig. 4: Correlation heatmap of selected crime-rate variables.

6.2 Multicollinearity Analysis using VIF

To further evaluate multicollinearity among the predictor variables, the Variance Inflation Factor (VIF) was calculated. In this analysis, *Murder_Rate* was treated as the target variable and was therefore not included in the VIF calculation. The VIF test was applied only to the remaining predictor variables: *Rape_Rate, Robbery_Rate, Aggravated_Assault_Rate, Larceny_Rate, Motor_Vehicle_Theft_Rate,* and *Burglary_Rate*.

Before feature removal, *Larceny_Rate* and *Burglary_Rate* had VIF values above 10, indicating high multicollinearity. Since these variables were strongly correlated, one highly collinear feature was removed. After feature removal, the VIF values of the remaining variables dropped to acceptable levels, all below 5. This indicates that the final predictor set was more stable and more suitable for clustering and classification analysis.

TABLE I: VIF Values Before Feature Removal

Feature	VIF
Rape_Rate	2.071
Robbery_Rate	4.324
Aggravated_Assault_Rate	5.623
Larceny_Rate	10.110
Motor_Vehicle_Theft_Rate	4.568
Burglary_Rate	10.188

TABLE II: VIF Values After Feature Removal

Feature	VIF
Rape_Rate	2.004
Robbery_Rate	4.080
Aggravated_Assault_Rate	4.780
Larceny_Rate	4.702
Motor_Vehicle_Theft_Rate	4.304

7 Model Building

7.1 Clustering

Clustering was used to group New York State counties according to similarities in crime behavior. Since the project focused on two main categories of crime, clustering was performed separately for violent crime and property crime. Because the selected variables were measured on different numerical scales, the data was standardized before applying K-Means clustering. The elbow method was used to determine the most appropriate number of groups, and in both the violent and property crime analyses, **k = 4** was selected.

7.1.1 Violent Crime Clustering

For violent crime clustering, the variables used were *Murder_Rate, Rape_Rate, Robbery_Rate,* and *Aggravated_Assault_Rate*. These variables were selected because they represent the major components of violent crime and provide a detailed view of county-level violent offense patterns. The cluster visualization using PCA showed that the counties formed separate groups based on their violent crime characteristics.

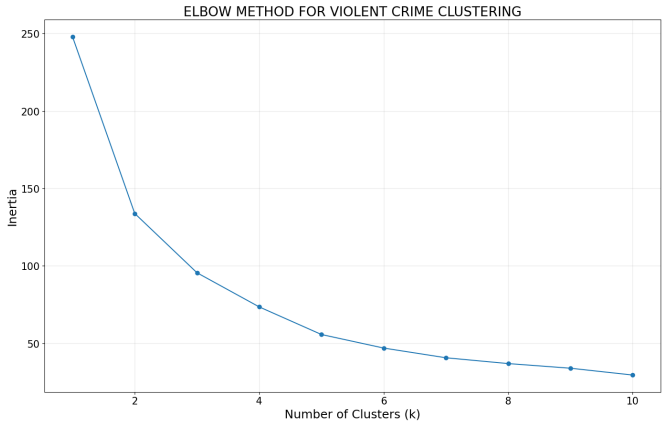

Fig. 5: Elbow method for violent crime clustering.

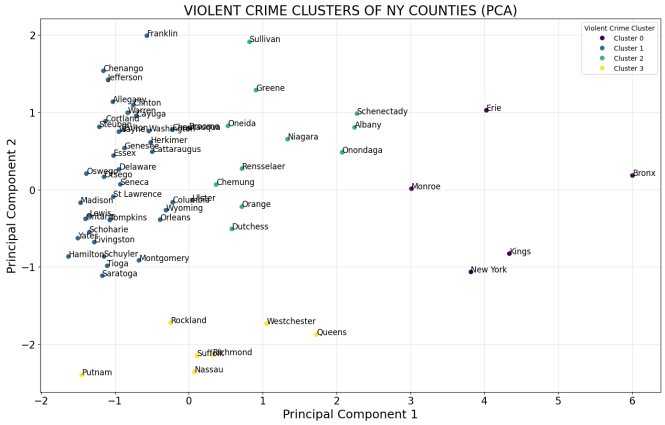

Fig. 6: Violent crime clustering with PCA-based visualization.

Although some overlap remained, the four clusters captured noticeable differences in violent crime structure across counties.

The scaled violent crime profile graph provides a clearer interpretation of these clusters by comparing the relative level of each violent crime variable within each group. The graph shows that **Cluster 0** had the highest overall violent crime intensity, with especially high values for murder, robbery, and aggravated assault. **Cluster 1** was mainly characterized by a very high rape rate, while the other violent crime variables remained relatively low. **Cluster 2** showed moderate to high levels across most violent crime variables, especially rape and aggravated assault, indicating a mixed but still serious violent crime pattern. **Cluster 3** had the lowest overall violent crime burden, with only a small contribution from murder and robbery and near-zero values for the other variables. This profile graph helps distinguish the crime characteristics of each cluster more clearly than the PCA plot alone. profile graph helps distinguish the crime characteristics of each cluster more clearly than the PCA plot alone.

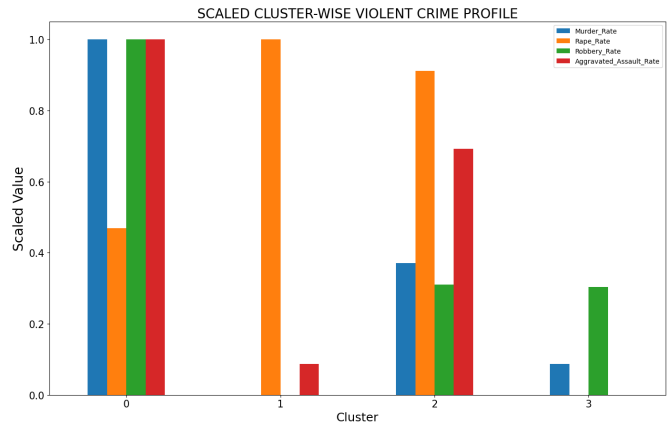

Fig. 7: Violent crime cluster-wise profile.

TABLE III: Violent Crime Cluster Profile

Cluster	Murder Rate	Rape Rate	Robbery Rate	Aggravated Assault Rate
0	12.14	38.99	439.30	523.88
1	3.05	62.70	31.43	257.53
2	6.42	58.76	157.86	434.31
3	3.84	17.98	155.34	232.29

7.1.2 Property Crime Clustering

For property crime clustering, the variables used were *Burglary_Rate, Larceny_Rate,* and *Motor_Vehicle_Theft_Rate*. These variables capture the main components of property crime and help reveal differences in county-level property offense structures. The PCA-based cluster visualization showed that the counties differed more strongly in some property crime measures than others, especially in motor vehicle theft and larceny. This indicates that not all counties follow the same type of property crime pattern.

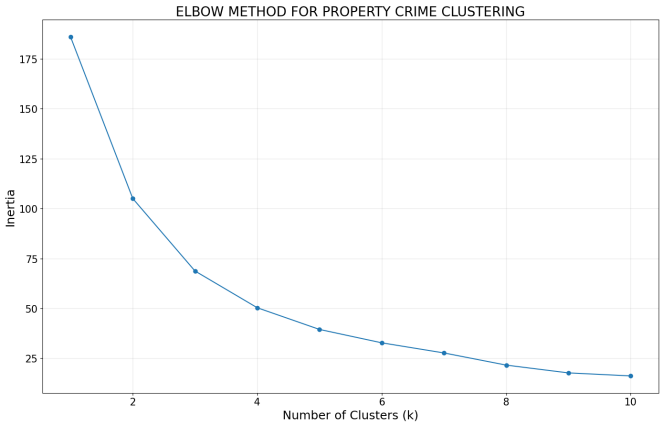

Fig. 8: Elbow method for property crime clustering.

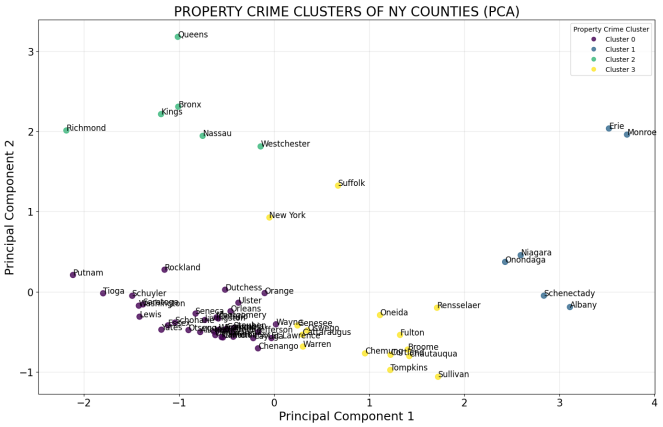

Fig. 9: Property crime clustering with PCA-based visualization.

The scaled property crime profile graph further explains the behavior of each cluster. **Cluster 1** showed the highest values for burglary, larceny, and motor vehicle theft, indicating the most severe overall property crime burden among the four groups. **Cluster 2** was mainly distinguished by a high motor vehicle theft rate, while burglary and larceny were much lower. **Cluster 3** showed moderately high burglary and

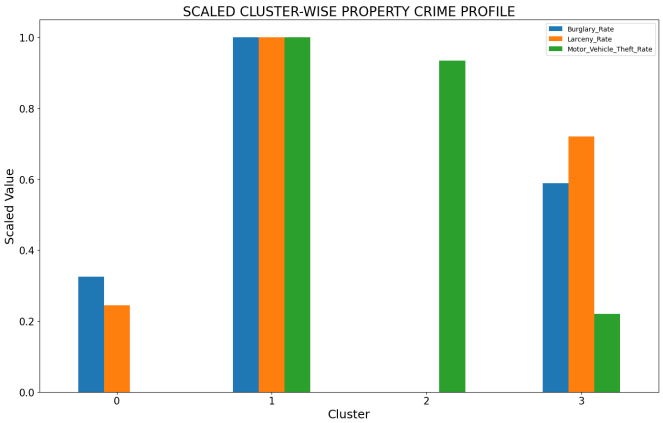

Fig. 10: Property crime cluster-wise profile.

larceny rates but a lower level of motor vehicle theft, representing a more balanced property crime pattern. **Cluster 0** had comparatively lower values across all three variables, indicating the least severe property crime profile.

TABLE IV: Property Crime Cluster Profile

Cluster	Burglary Rate	Larceny Rate	Motor Vehicle Theft Rate
0	716	2391	101
1	1270	4501	481
2	450	1707	457
3	934	3721	185

7.1.3 PCA for Cluster Visualization and Interpretation

Principal Component Analysis (PCA) was used within the clustering framework to support visualization and interpretation of the county groups. For the violent crime analysis,**91.12%** of the total variance was explained by the first two principal components. For the property crime analysis, **89%** of the total variance was explained by the first two principal components. This indicates that most of the variation present in the original crime variables was retained even after reducing the data to two dimensions.

The PCA plots made it easier to visually examine how counties were distributed across the identified clusters. Counties that appeared close together in the PCA space had more similar crime patterns, while those located farther apart showed greater differences in crime characteristics. In the violent crime PCA plot, a few counties appeared more clearly separated from the rest, suggesting stronger violent crime intensity or a distinct combination of violent crime features. In contrast, several counties remained grouped near the center, indicating similar violent crime behavior. A similar pattern was observed in the property crime PCA plot, where some clusters were more compact while others showed moderate overlap. This suggests that although certain county groups shared related crime structures, the clustering still captured meaningful differences among them.

Overall, PCA improved the interpretability of the clustering results by simplifying the high-dimensional crime data into a smaller visual space without losing much information. It also helped confirm that the identified clusters reflected meaningful county-level differences in violent and property crime patterns across New York State.

7.2 Logistic Regression

The Logistic Regression results indicate that the relationship between the predictor variables and the crime categories can be reasonably captured using a linear decision boundary in the transformed feature space. Compared with the Decision Tree and Random Forest models, Logistic Regression provided the most balanced overall performance across the three crime classes. The model performed particularly well for the **Low** crime category, while some confusion remained between the

High and **Medium** classes, suggesting partial overlap in their feature patterns.

The performance of Logistic Regression also shows that the selected predictors were informative enough to distinguish the three crime classes with moderate success. Since Logistic Regression is a simpler and more interpretable model than the tree-based methods, its strong performance suggests that the classification problem did not require very complex nonlinear boundaries to achieve reasonable separation. This makes the model useful not only for prediction but also for understanding which variables contributed most to the classification process. The relatively close values of precision, recall, and F1-score further indicate that the model maintained a stable balance between identifying the classes correctly and avoiding excessive misclassification.

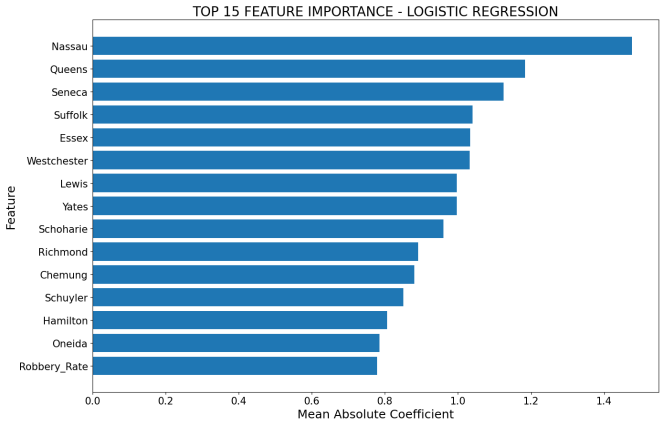

Fig. 11: Top 15 feature importance values for Logistic Regression based on mean absolute coefficient magnitude.

The feature importance analysis further showed that both county-specific variables and crime-rate variables contributed to the classification process. Counties such as *Nassau, Queens, Suffolk,* and *Westchester* appeared among the most influential predictors, indicating that geographic location played an important role in distinguishing crime levels. Among the numerical crime-related variables, *Robbery_Rate* was one of the strongest contributors, showing that robbery-related variation had a strong influence on class separation. This result is important because it suggests that both regional characteristics and offense-specific crime rates influenced the final classification outcome. In particular, the prominence of robbery-related information indicates that this variable carried substantial discriminatory power across the **High, Low,** and **Medium** classes. Overall, the Logistic Regression model served as a strong and interpretable baseline and achieved the best overall predictive performance among the three classification models.

TABLE V: Logistic Regression Classification Report

Class	Precision	Recall	F1-score	Support
High	0.78	0.53	0.63	144
Low	0.59	0.77	0.67	145
Medium	0.59	0.60	0.60	145
Accuracy		0.63		434
Macro Avg	0.65	0.63	0.63	434
Weighted Avg	0.65	0.63	0.63	434

TABLE VI: Logistic Regression Feature Importance

Feature	Importance
Nassau	1.475922
Queens	1.184122
Seneca	1.125177
Suffolk	1.040519
Essex	1.033567
Westchester	1.033057
Lewis	0.996771
Yates	0.996089
Schoharie	0.960532
Richmond	0.891269
Chemung	0.880339
Schuyler	0.850308
Hamilton	0.805372
Oneida	0.783892
Robbery_Rate	0.778902

7.3 Decision Tree

A Decision Tree classifier was developed to classify observations into **High, Low,** and **Medium** crime categories. Decision Trees are useful because they are easy to interpret and can capture nonlinear relationships through repeated feature-based splits.

The best cross-validation score was **0.5446**. On the test set, the model achieved an **accuracy of 0.5853**, **precision of 0.6077**, **recall of 0.5853**, and **F1-score of 0.5822**. These results show that the Decision Tree classified the crime categories with moderate performance, although its predictive accuracy was lower than both Logistic Regression and Random Forest. The model performed better for the **Low** crime category, while the **High** and **Medium** classes showed greater overlap and were more difficult to separate correctly. This suggests that although the tree captured meaningful classification rules, the simple tree structure was not sufficient to fully distinguish all three crime levels.

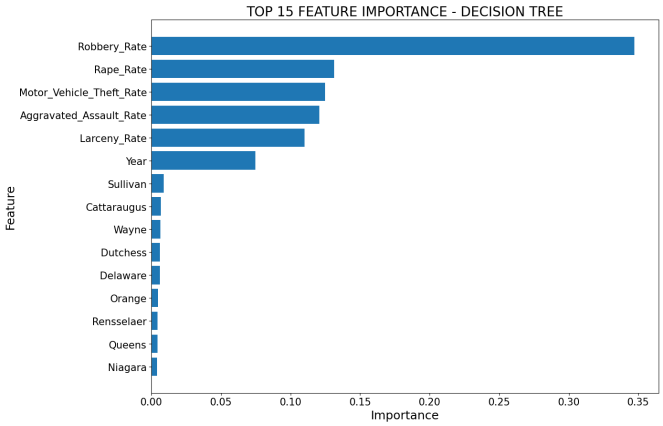

Fig. 12: Top 15 feature importance values for Decision Tree.

One of the main advantages of the Decision Tree model is its interpretability. The tree visualization clearly shows how the model used repeated splits on important predictors to classify the observations. In particular, *Robbery_Rate* appeared as the most dominant splitting variable and also ranked as the most important feature overall. Other variables such as *Rape_Rate, Motor_Vehicle_Theft_Rate, Aggravated_Assault_Rate, Larceny_Rate,* and *Year* also contributed to the classification process, indicating that both violent and property crime characteristics influenced the final class separation. Overall, although the Decision Tree did not achieve the highest predictive performance, it provided a simple and interpretable structure for understanding how the crime categories were differentiated.

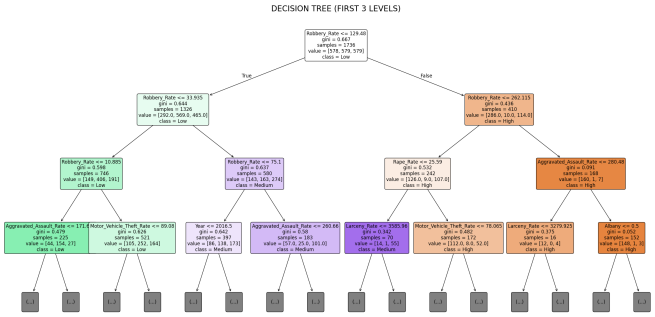

Fig. 13: Decision Tree visualization.

TABLE VII: Decision Tree Classification Report

Class	Precision	Recall	F1-score	Support
High	0.74	0.48	0.58	144
Low	0.55	0.74	0.63	145
Medium	0.53	0.53	0.53	145
Accuracy		0.59		434
Macro Avg	0.61	0.59	0.58	434
Weighted Avg	0.61	0.59	0.58	434

TABLE VIII: Decision Tree Feature Importance

Feature	Importance
Robbery_Rate	0.903486
Rape_Rate	0.090331
Aggravated_Assault_Rate	0.006183
Albany	0.000000
Richmond	0.000000
Schuyler	0.000000
Schoharie	0.000000
Schenectady	0.000000
Saratoga	0.000000
Rockland	0.000000
Queens	0.000000
Rensselaer	0.000000
St Lawrence	0.000000
Putnam	0.000000
Otsego	0.000000

7.4 Random Forest

Random Forest was used as an ensemble classification model made up of multiple decision trees. It was chosen because ensemble learning generally improves predictive performance and reduces overfitting compared with a single Decision Tree.

The best Random Forest model used **100 trees**, **max_features = log2**, **min_samples_split = 5**, **min_samples_leaf = 2**, and **max_depth = None**. The best cross-validation score was **0.5898**. On the test set, the model achieved an **accuracy of 0.6221**, **precision of 0.6332**, **recall of 0.6221**, and **F1-score of 0.6187**.

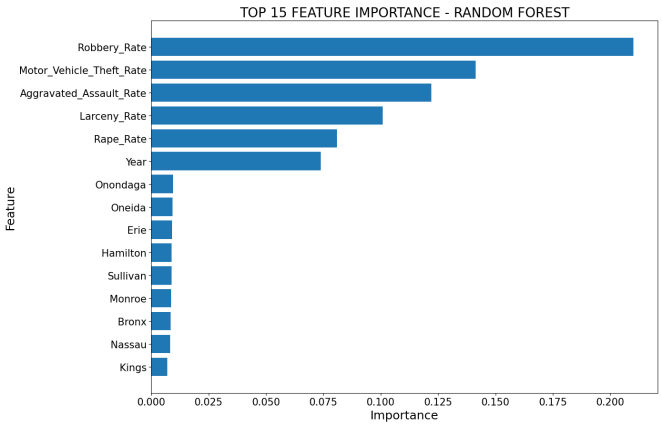

Fig. 14: Top 15 feature importance values for Random Forest.

These results show that Random Forest performed better than the single Decision Tree and slightly below Logistic Regression. The improvement over the Decision Tree suggests that combining multiple trees helped produce a more stable and reliable classification model. Similar to the other models, the **Low** crime category was classified more accurately, while the **High** and **Medium** categories were more difficult to separate because of overlap in their predictor patterns.

One important advantage of Random Forest is that it can capture more complex relationships in the data

while also reducing the risk of overfitting that often affects a single tree. The feature importance results showed that *Robbery_Rate* was the most important predictor, followed by *Motor_Vehicle_Theft_Rate, Aggravated_Assault_Rate, Larceny_Rate, Rape_Rate,* and *Year*. This indicates that both violent crime and property crime variables contributed meaningfully to the classification process. In addition, several county-specific variables such as *Onondaga, Oneida, Erie, Hamilton, Sullivan, Monroe, Bronx, Nassau,* and *Kings* also appeared among the top features, suggesting that geographic effects also played a role in distinguishing crime categories. Overall, Random Forest provided strong predictive performance and useful feature importance insights, even though Logistic Regression remained the best-performing model overall.

TABLE IX: Random Forest Classification Report

Class	Precision	Recall	F1-score	Support
High	0.72	0.54	0.62	144
Low	0.59	0.78	0.67	145
Medium	0.59	0.54	0.57	145
Accuracy	0.62			434
Macro Avg	0.63	0.62	0.62	434
Weighted Avg	0.63	0.62	0.62	434

TABLE X: Random Forest Feature Importance

Feature	Importance
Robbery_Rate	0.210185
Motor_Vehicle_Theft_Rate	0.141402
Aggravated_Assault_Rate	0.122004
Larceny_Rate	0.100808
Rape_Rate	0.080960
Year	0.073838
Onondaga	0.009471
Oneida	0.009211
Erie	0.009112
Hamilton	0.008971
Sullivan	0.008806
Monroe	0.008603
Bronx	0.008408
Nassau	0.008191
Kings	0.006969

8 Results

The classification results show that all three models—Logistic Regression, Decision Tree, and Random Forest—were able to classify observations into **High, Low,** and **Medium** crime categories with moderate performance. However, their effectiveness differed in terms of overall accuracy, class-wise prediction, interpretability, and feature usage.

Among the three models, **Logistic Regression** achieved the best overall performance. It produced a test accuracy of **0.6336**, with **precision = 0.6530**, **recall = 0.6336**, and **F1-score = 0.6316**. The model performed best for the **Low** class, while the **High** and **Medium** classes showed greater overlap. This suggests that the lower crime observations had more clearly identifiable patterns, whereas the higher and medium crime groups shared more similar characteristics. Logistic Regression also provided the most balanced results across the evaluation metrics, which indicates that it was able to classify the observations more consistently than the other two models.

The **Decision Tree** model produced lower performance compared with Logistic Regression. It achieved a test accuracy of **0.5853**, with **precision = 0.6077**, **recall = 0.5853**, and **F1-score = 0.5822**. The model predicted the **Low** class better than the other two classes, while **High** and **Medium** showed more misclassification. Although its predictive performance was lower, the Decision Tree had an important advantage in terms of interpretability. The tree structure clearly illustrated how the model used repeated feature-based splits to separate the three classes. This made it easier to understand which variables influenced the prediction process and how the classes were differentiated step by step.

The **Random Forest** model performed better than the Decision Tree and slightly below Logistic Regression. It achieved a test accuracy of **0.6221**, with **precision = 0.6332**, **recall = 0.6221**, and **F1-score = 0.6187**. Like the other models, it performed best for the **Low** class, while **High** and **Medium** remained harder to separate. The improved performance over the single Decision Tree suggests that the ensemble structure of Random Forest helped reduce instability and improved generalization. By combining many trees, the model captured more complex patterns in the crime data and produced more reliable classification results.

A comparison of feature importance across the three models showed several consistent findings. *Robbery_Rate* appeared as one of the most influential predictors in all classification models. It was included among the important variables in Logistic Regression, dominated the Decision Tree structure, and ranked first in Random Forest. In addition, other crime-related variables such as *Motor_Vehicle_Theft_Rate, Aggravated_Assault_Rate, Larceny_Rate, Rape_Rate,* and *Year* also contributed to classification, especially in the tree-based models. County-specific variables such as *Nassau, Queens, Suffolk, Westchester, Onondaga, Erie, Monroe, Bronx,* and *Kings* also appeared among the important features, suggesting that both geographic location and crime-type variation influenced class prediction.

Overall, the comparison of the three models shows that **Logistic Regression** performed best, followed closely by **Random Forest**, while **Decision Tree** gave the lowest predictive accuracy. Even though all three models achieved only moderate classification performance, they still provided meaningful insights into the structure of county-level crime patterns. The classification results indicate that the crime categories were distinguishable to a reasonable extent, but not perfectly separable. This is expected in a real-world dataset where counties may share overlapping social, demographic, and crime-related characteristics.

9 Conclusion

This project analyzed county-level crime patterns in New York State using historical crime and population data. By aggregating agency-level crime records to the county-year level and converting raw counts into population-adjusted crime rates, the study created a fair basis for comparing counties with different population sizes. This preprocessing step was important because it transformed the raw data into a more meaningful form for both clustering and classification analysis.

The feature significance analysis showed that several crime-rate variables were related to one another and that multicollinearity needed to be addressed before model development. The correlation analysis revealed moderate to strong positive relationships among some predictors, while the VIF analysis confirmed that certain variables carried overlapping information. After feature removal, the predictor set became more stable and more suitable for downstream analysis. This step improved the reliability of the clustering and classification procedures.

The clustering analysis showed that both violent and property crime patterns could be meaningfully divided into four county groups. The violent crime clustering highlighted clear differences in the relative contribution of murder, rape, robbery, and aggravated assault across counties, while the property crime clustering revealed different patterns in burglary, larceny, and motor vehicle theft. The scaled crime profile graphs helped interpret the character of each cluster more clearly by showing which crime variables were relatively dominant within each group. This made the clusters easier to describe and compare in a practical way.

PCA further supported the clustering analysis by reducing the dimensionality of the data while still retaining most of the total variation. The first two principal components explained a large share of the variance in both the violent and property crime analyses, which made it possible to visualize county groupings in a lower-dimensional space. Although some overlap remained between clusters, the PCA plots still confirmed that the cluster structure captured meaningful differences in county-level crime behavior.

The classification analysis showed that Logistic Regression, Decision Tree, and Random Forest could classify crime-level categories with moderate success, with Logistic Regression achieving the best overall performance. The Decision Tree provided the most interpretable structure, while Random Forest improved predictive performance by combining multiple trees. Across all classification models, *Robbery_Rate* consistently emerged as one of the most influential predictors. This suggests that robbery-related variation plays an especially important role in distinguishing county-level crime categories in this dataset.

Overall, this study demonstrates that data mining methods such as correlation analysis, VIF, K-Means clustering, PCA, Logistic Regression, Decision Tree, and Random Forest can effectively reveal meaningful county-level crime structures in New York State. The results show that both crime-type variables and county-specific characteristics contribute to the observed patterns. While the models achieved only moderate predictive performance, they still provided useful analytical insights into how counties differ in violent and property crime behavior.

This study can be extended further in future work by incorporating additional explanatory variables such as socioeconomic indicators, unemployment levels, education, or population density. Including such variables may improve the interpretation of the clusters and strengthen the predictive performance of the classification models. Time-series forecasting methods could also be explored in greater depth to examine how county-level crime patterns may change in future years. Even with its current scope, the project provides a strong foundation for understanding county-level crime variation in New York State through data mining techniques.

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